

Numerical Simulation of Transient Vibrational Power Flows in Slender Heterogeneous Structures

É. Savin

Aeroelasticity and Structural Dynamics Department
ONERA, Châtillon, France

Abstract

This paper deals with some recent developments for the modeling and numerical simulation of high-frequency (HF) vibrations of heterogeneous slender structures. The mathematical model is derived from the semiclassical analysis of strongly oscillating (HF) solutions of quantum and classical wave systems, including acoustic, electromagnetic, or in the present case elastic waves. This theory shows that the associated phase-space energy density satisfies a transport equation in the HF range. The proposed model also considers energetic boundary and interface conditions consistent with the boundary and interface conditions imposed to the solutions of the wave system. They are given in the form of power flow reflection/transmission operators for the energy rays impinging on a boundary or an interface. Specular-like transverse boundary reflections, diffuse reflections, or fluid-structure coupling may be treated as a particular case of the proposed model. Nodal/spectral discontinuous 'Galerkin' finite element methods and Monte-Carlo methods are implemented to integrate the transport equations with such boundary and interface conditions. Some numerical simulations are presented to illustrate the theory: the first one deals with an assembly of thick beams, and the second one with an assembly of thick shells. This research applies to the prediction of the linear transient responses of complex structures to impact loads or shocks, or the analysis of non-destructive evaluation techniques.

Keywords: high-frequency, vibration, transport, discontinuous finite elements, Monte-Carlo method.

1 Introduction

Engineering structures exhibit typical transport and diffusive behaviors in the higher frequency (HF) range of vibration [22]. These regimes are described by linear trans-

port equations for the energy density associated to the strongly oscillating (HF) solutions of the Navier-Cauchy equation for elastic wave propagation. More generally this result holds for all symmetric hyperbolic systems, including quantum waves and classical acoustic or electromagnetic waves [2, 3, 13, 20]. It has been specialized to slender visco-elastic structures, typically beams, plates and shells in [23, 24, 26] for applications to aerospace structures. The mathematical model is derived from the semiclassical analysis of HF solutions of such wave systems: it shows that the associated energy density in the phase-space position $\times$ wave vector satisfies a transport equation in the HF range. The transport equation allows to track the energy paths (rays) within the propagation medium, thus it also describes all power flows in slender structures. This theory provides a rational framework for the validation, and generalization, of heuristic methods such as the statistical energy analysis (SEA, see [18]) or the vibrational conductivity analogy (VCA, see [19]). Its main difficulty to date lies in the consideration of boundary and interface conditions for quadratic quantities, e.g. the energy and power flow densities, consistent with the boundary and interface conditions imposed to the displacement and stress fields of the original elastic wave system. The former are given as power flow reflection/transmission operators for the energy rays impinging on a boundary or an interface. Specular-like transverse boundary reflections, diffuse reflections, or fluid-structure coupling may be treated as a particular case. A formal derivation of these HF operators at interfaces between slender sub-structures such as beams or shells has been proposed in [26]. More recently, a systematic derivation of the elastic energy density boundary conditions near the doubly-hyperbolic set (where both compressional and shear elastic waves are incident on the boundary) and the hyperbolic-elliptic set (where shear waves are incident on the boundary with an angle greater than their critical angle) has been proposed in [2]. The analysis however does not account for the glancing set (at critical incidence) and the transport of energy along an interface by so-called gliding rays; this issue is the subject of ongoing research. Yet a transport model in bounded media with general transverse or diffuse boundary conditions for the power flows may be derived from these results [27].

Transport equations can be solved numerically by various methods [9, 11], including Monte-Carlo methods [16] or nodal/spectral discontinuous finite element methods [5, 6, 15, 17, 21]. Both techniques offer the great advantage of being numerically weakly dissipative and weakly dispersive depending on the order of interpolation. Thus they are well suited for long-time simulations, when transport equations in bounded media converge to diffusion equations [8, 28]. The diffusive regime is characterized by energy equilibration rules which can be exhibited from direct numerical simulations using those schemes [27]. Its onset at late times is prompted by the multiple scattering of energy rays on boundaries and interfaces. The diffusive regime is precisely the one invoked in the SEA and VCA approaches to derive steady-state global or local diffusion equation by heuristic arguments. The main purpose of this paper is to outline the transport theory actually yielding diffusion for high-frequency structural vibrations, and more specifically the numerical methods which may be used to solve transient transport equations in view of demonstrating their diffusive behavior

at late times. It is organised as follows. First, the transport model for the evolution of the high-frequency energy density in heterogeneous structures is resumed, considering boundary and interface conditions as well. The numerical methods implemented for solving the corresponding transport equations are detailed in Sect. 3. Then two numerical examples are introduced in Sect. 4: the first one deals with an assembly of thick beams, and the second one with an assembly of thick shells. At last Sect. 5 offers some conclusions and directions of future works.

2 Transport model

In Sect. 2.1 one summarizes the basic results obtained in [2, 3, 13, 20] for the transport properties of high-frequency waves in elastic media. The boundary conditions to be added to the evolution model recalled below are detailed in Sect. 2.2 for specular-like or diffusive-like reflections at outer boundaries and in Sect. 2.3 for reflections and transmissions at interfaces.

2.1 Transport in an open domain

Let us consider an heterogeneous, visco-elastic medium embedded in the open domain $\mathcal{O} \subseteq \mathbb{R}^d$, where $\mathbb{R}$ is the set of all real numbers and $d = 1, 2$ or 3 is the physical space dimension. Then the evolution of the high-frequency energy density associated to high-frequency waves propagating in $\mathcal{O}$ follows the multigroup transport equations (MTE) [2, 3, 13, 20]:

$$\partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} = 0, \quad (1)$$

which couple M different propagation modes, or energy rays, of intensity w_α for $1 \leq \alpha \leq M$. Typically $M = 3$ and $\alpha = \text{P, SV or SH}$ in an isotropic elastic medium, for the compressional and shear waves. Here $\lambda_\alpha(\mathbf{x}, \mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ is the eigenfrequency of the mode α of which group velocity is c_α , $\{f, g\} = \nabla_{\mathbf{k}} f \cdot \nabla_{\mathbf{x}} g - \nabla_{\mathbf{x}} f \cdot \nabla_{\mathbf{k}} g$ is the usual Poisson's bracket, and $\mathbb{S}^{d-1}$ is the unit sphere of $\mathbb{R}^d$ with the uniform probability measure μ . The scalar intensities $\{w_\alpha\}_{1 \leq \alpha \leq M}$, hereafter called phase-space energy densities or specific intensities, are non-negative if the initial data are non-negative. Then the space-time energy density $\mathcal{E}(\mathbf{x}, t) \in \mathbb{R}^+$ and power flow density $\Pi(\mathbf{x}, t) \in \mathbb{R}^d$ of this medium are recovered by:

$$\mathcal{E}(\mathbf{x}, t) = \frac{1}{2} \sum_{\alpha=1}^M \int_{\mathbb{R}^d} w_\alpha(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}, \quad \Pi(\mathbf{x}, t) = \frac{1}{2} \sum_{\alpha=1}^M c_\alpha(\mathbf{x}) \int_{\mathbb{R}^d} w_\alpha(\mathbf{x}, \mathbf{k}, t) \hat{\mathbf{k}} d\mathbf{k}. \quad (2)$$

These results are adapted to elastic waves in slender structures in [23, 24, 26], but they also apply to waves in poro-visco-elastic media, acoustic waves, electromagnetic waves, or the Schrödinger equation in random media [20].

For convenience, the MTE of Equation (1) are reshaped in the form of a system of first-order conservative equations (3) in phase space $T^*\mathcal{O} := \mathcal{O} \times \mathbb{R}^d$ as follows.

Let us introduce the phase variable $\mathbf{z} = (\mathbf{x}, \mathbf{k}) \in T^*\mathcal{O}$, the set $E = \{1, 2, \dots, M\}$, the vector $\mathbf{W} = (w_1, w_2, \dots, w_M)^T$ of $(\mathbb{R}_+)^M$, and the linear flux operator:

$$\mathcal{F}(\mathbf{W}) = (w_1 \mathbf{F}_1, w_2 \mathbf{F}_2, \dots, w_M \mathbf{F}_M)^T, \quad \mathbf{F}_\alpha(\mathbf{z}) = \begin{pmatrix} c_\alpha(\mathbf{x}) \hat{\mathbf{k}} \\ -|\mathbf{k}| \nabla_{\mathbf{x}} c_\alpha \end{pmatrix}$$

($\mathcal{F}(\mathbf{W})$ is thus a $M \times 2d$ real matrix). Then Equation (1) writes:

$$\partial_t \mathbf{W} + \text{Div}_{\mathbf{z}} \mathcal{F}(\mathbf{W}) = \mathbf{0}, \quad (3)$$

which is a conservative system. On any discontinuity front Σ_D in phase-space defined by the hypersurface $\Sigma_D = \{\mathbf{z} \in T^*\mathcal{O}; S(\mathbf{z}) = 0\}$, its piecewise continuous (weak) solution $\mathbf{W}$ satisfies the Rankine-Hugoniot condition:

$$\sum_{\alpha \in E} [\![S, \lambda_\alpha]\!] w_\alpha = 0,$$

where $[\![f]\!]$ stands for the jump of f on the front Σ_D . This relation states that the total normal power flow density $\sum_{\alpha \in E} c_\alpha w_\alpha (\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_\Sigma)$ remains continuous accross that front, denoting by $\hat{\mathbf{n}}_\Sigma$ its normal; in return the specific intensities themselves have no reason to remain continuous. The Rankine-Hugoniot condition does not describe how the individual normal power flows $c_\alpha w_\alpha (\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_\Sigma)$ for each mode $\alpha \in E$ are distributed inward and outward on the front. This issue is considered in the next two sections for Σ_D being either a boundary or an interface (a junction) where the group velocities in the medium are discontinuous.

2.2 Transport in a bounded domain

Now the above system (3), completed by square-integrable initial conditions $\mathbf{W}(\mathbf{z}, 0) = \mathbf{W}^0(\mathbf{z})$, is considered in the phase space $T^*\mathcal{D} = \mathcal{D} \times \mathbb{R}^d$ where $\mathcal{D}$ is a bounded subset of $\mathcal{O}$, and for $t \in [0, T]$, $T < +\infty$ but arbitrary. It is assumed that $\mathbf{W}^0(\mathbf{z}, \{\mathbf{k} = \mathbf{0}\}) = \mathbf{0}$. Wavenumber $|\mathbf{k}| > 0$ is a fixed parameter, and we shall also assume as usual that the velocities $c_\alpha \in W^{1,\infty}(\mathcal{D})$ are positive functions with possible isolated discontinuities if $\mathcal{D}$ contains sharp interfaces between different materials. For any smooth manifold Σ of co-dimension 1 in $\mathbb{R}^d$ oriented by its normal $\hat{\mathbf{n}}_\Sigma$, let:

$$\Gamma^\pm(\Sigma) = \{(\mathbf{x}, \mathbf{k}) \in \Sigma \times \mathbb{R}^d; \pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_\Sigma(\mathbf{x}) > 0\}.$$

Then well-posedness of the problem is ensured by setting proper boundary conditions on the inflow boundary $\Gamma^-(\partial\mathcal{D})$ of the computational domain $\mathcal{D}$ [4]. Denoting by $\hat{\mathbf{n}}$ the outward unit normal to the (smooth) boundary $\partial\mathcal{D}$, an incoming flux $\mathbf{f}(\mathbf{z}, t)$ can be imposed as $-\mathcal{F}(\mathbf{W})\hat{\mathbf{n}} = \mathbf{f}$ on $\Gamma^-(\partial\mathcal{D})$ and for $t > 0$. However generalized boundary reflection laws can also be considered in the form:

$$-\mathcal{F}(\mathbf{W})\hat{\mathbf{n}} = \mathbf{f} + \mathcal{R}_{\mathbf{x}}(\mathbf{W}), \quad \mathbf{z} \in \Gamma^-(\partial\mathcal{D}), \quad t > 0, \quad (4)$$

where $\mathcal{R}_x$ is a boundary reflection operator defined from $\Gamma^+(\partial\mathcal{D})$ into $\Gamma^-(\partial\mathcal{D})$. Note that it depends only on the *local* geometry of the boundary (it may also depend on $|\mathbf{k}|$). This operator describes how the outward normal power flows $\mathcal{F}(\mathbf{W})\hat{\mathbf{n}}$, $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}} > 0$, are reflected into inward normal power flows $-\mathcal{F}(\mathbf{W})\hat{\mathbf{n}}$, $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}} < 0$, with possible mode conversions in E . Here the glancing set $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}} = 0$ is ignored, not because such grazing rays are unlikely to occur, but because we lack a theoretical model to describe the associated boundary specific intensities at present. The rigorous construction of $\mathcal{R}_x$ away from the glancing set is detailed in [2] for elastic waves. A formal derivation for bending, compressional and shear waves in thick beams and shells is given in [26]. Its basic properties and other examples are considered in [27].

2.3 Transport with a sharp interface

Assume that the physical domain $\mathcal{D}$ is partitioned into non-overlapping sub-domains and their interface $\gamma_D \subset \overline{\mathcal{D}}$ is a smooth, bounded manifold of codimension 1 oriented by its unit normal $\hat{\mathbf{n}}_D$ (ignoring edges and corners at a first glance). More specifically, γ_D is any interface between these sub-domains or between a sub-domain and the exterior of $\mathcal{D}$ and consequently, it may also be a part or the whole of $\partial\mathcal{D}$. If the group velocities c_α are continuous across γ_D , the continuity of the normal power flow on γ_D is ensured by the conditions $\mathcal{F}(\mathbf{W}^+)\hat{\mathbf{n}}_D = \mathcal{F}(\mathbf{W}^-)\hat{\mathbf{n}}_D$ if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D > 0$ (forward flux) and $-\mathcal{F}(\mathbf{W}^-)\hat{\mathbf{n}}_D = -\mathcal{F}(\mathbf{W}^+)\hat{\mathbf{n}}_D$ if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D < 0$ (backward flux), where $\mathbf{W}^\pm(\mathbf{x}, \mathbf{k}, t) = \lim_{h \downarrow 0} \mathbf{W}(\mathbf{x} \pm h\hat{\mathbf{n}}_D, \mathbf{k}, t)$ for a.e. $\mathbf{x} \in \gamma_D$, and $\mathbf{W}^+(\mathbf{x}, \mathbf{k}, t) = \mathbf{0}$ whenever $\mathbf{x} \in \partial\mathcal{D}$; likewise, $c_\alpha^\pm(\mathbf{x}) = \lim_{h \downarrow 0} c_\alpha(\mathbf{x} \pm h\hat{\mathbf{n}}_D)$ a.e. $\mathbf{x} \in \gamma_D$. If however the group velocities are discontinuous across γ_D , such that $c_\alpha^+ \neq c_\alpha^-$, these expressions may be generalized to the form:

$$\begin{aligned} \mathcal{F}(\mathbf{W}^+)\hat{\mathbf{n}}_D &= \mathcal{R}_D^+(\mathbf{W}^+) + \mathcal{T}_D^-(\mathbf{W}^-), & \mathbf{z} \in \Gamma^+(\gamma_D), \\ -\mathcal{F}(\mathbf{W}^-)\hat{\mathbf{n}}_D &= \mathcal{R}_D^-(\mathbf{W}^-) + \mathcal{T}_D^+(\mathbf{W}^+), & \mathbf{z} \in \Gamma^-(\gamma_D), \end{aligned} \quad (5)$$

$t > 0$, where $\mathcal{R}_D^\pm : \Gamma^\mp(\gamma_D) \rightarrow \Gamma^\pm(\gamma_D)$ and $\mathcal{T}_D^\mp : \Gamma^\pm(\gamma_D) \rightarrow \Gamma^\pm(\gamma_D)$ are the upstream (−) and downstream (+) reflexion and transmission operators, respectively, of the interface γ_D . These relations show how the inward and outward normal power flows are distributed on the interface: the outward flows are the sum of reflected and transmitted flows, taking into account possible mode conversions in E , on either side of that interface. The explicit derivation of the reflection and transmission operators for junctions of thick beams or shells is outlined in [26]. Their general properties and some examples are also given in [27].

3 Numerical methods

Numerical methods with low numerical dispersion and dissipation errors are needed to perform long-time simulations of the MTE in order to reach its diffusion limit. Monte-Carlo methods and discontinuous finite element methods both have these properties.

Monte-Carlo methods are very easy to implement since they are based on an intuitive interpretation of transport equations. They also allow to compute local solutions, a decisive advantage over energetic methods (such as finite elements) for large-scale computations. They converge slowly, but it is always possible to control their accuracy. Their main drawback is their lack of versatility for complex geometries as typically encountered in structural dynamics and acoustics. Finite element methods are, on the contrary, much more flexible and can be applied to truly complex geometries. Among these methods, the discontinuous "Galerkin" (DG) finite element method has originally been introduced by Reed & Hill [21] and Lesaint & Raviart [17] to compute neutronic transfers; it is therefore adapted to the integration of MTEs. In the DG method, boundary fluxes at the edges or faces of the elements maintain the consistency of the numerical solution with the continuous, non-discretized MTEs, which are otherwise discretized on each element without any continuity relation between elements. This scheme is recalled in Sect. 3.2 below; the numerical fluxes are constructed in order to get to satisfy the boundary and interface conditions (4) and (5).

3.1 Monte-Carlo method

The Monte-Carlo method for solving a transport partial differential equation is based on its interpretation as a Fokker-Planck equation for the marginal probability density $\rho = (\rho_\alpha)_{\alpha \in E}$ of an underlying homogeneous Markov process $\{(\mathbf{Z}_t, A_t), t \geq 0\}$ having its values in $T^*\mathbb{R}^d \times E$ [16]. Let $\mathcal{A}$ be the infinitesimal generator of that process, that is:

$$\mathcal{A}\phi = \lim_{h \downarrow 0} \frac{1}{h} (\mathbb{E}_{\mathbf{z}, \alpha} \{\phi(\mathbf{Z}_h, A_h)\} - \phi(\mathbf{z}, \alpha)) , \quad \forall \phi \in \mathcal{C}_b^1(T^*\mathbb{R}^d \times E) ,$$

where $\mathbb{E}_{\mathbf{z}, \alpha} \{\phi(\mathbf{Z}_t, A_t)\} := \mathbb{E}\{\phi(\mathbf{Z}_t, A_t) | (\mathbf{Z}_t, A_t)_{t=0} = (\mathbf{z}, \alpha)\}$, $\mathbb{E}\{\cdot\}$ stands for the mathematical expectation, and $\mathcal{C}_b^1(T^*\mathbb{R}^d \times E)$ is the set of bounded functions which are $\mathcal{C}^1$ with respect to the physical space variable $\mathbf{x} \in \mathbb{R}^d$. The associated Fokker-Planck, or forward Kolmogorov equation is $\partial_t \rho(\mathbf{z}, t) = \mathcal{A}^* \rho(\mathbf{z}, t)$, where $\mathcal{A}^*$ stands for the adjoint operator. This is exactly the MTE (3) if $\mathcal{A}$ is chosen as $\mathcal{A}\rho := (\mathbf{F}_\alpha \cdot \nabla_{\mathbf{z}} \rho_\alpha)_{\alpha \in E}$. Its solution ρ satisfies:

$$\sum_{\alpha \in E} \int_{T^*\mathbb{R}^d} \phi(\mathbf{z}, \alpha) \rho_\alpha(d\mathbf{z}, t) = \sum_{\alpha \in E} \int_{T^*\mathbb{R}^d} \mathbb{E}_{\mathbf{z}, \alpha} \{\phi(\mathbf{Z}_t, A_t)\} \rho_\alpha^0(d\mathbf{z}) , \quad \forall \phi \in \mathcal{C}_b^1(T^*\mathbb{R}^d \times E) ,$$

starting with an initial condition $\rho(\mathbf{z}, 0) = \rho^0(\mathbf{z})$. Then taking formally $\phi(\mathbf{z}, \alpha) = \delta(\mathbf{z} - \mathbf{z}_0) \otimes \delta(\alpha - \alpha_0)$ (but $\phi \notin \mathcal{C}_b^1(T^*\mathbb{R}^d \times E)$!) and the usual estimator of the mathematical expectation $\mathbb{E}\{\phi(\mathbf{Z}_t, A_t)\} \simeq \frac{1}{N} \sum_{j=1}^N \phi(\mathbf{z}^j(t), \alpha^j(t))$ constructed by simulating N trajectories $t \mapsto (\mathbf{z}^j(t), \alpha^j(t))$, $1 \leq j \leq N$, of the process $\{(\mathbf{Z}_t, A_t), t \geq 0\}$, yields:

$$w_{\alpha_0}(\mathbf{z}_0, t) \simeq \frac{\mathcal{E}_0}{N} \sum_{j=1}^N \delta(\mathbf{z}^j(t) - \mathbf{z}_0) \otimes \delta(\alpha^j(t) - \alpha_0) ,$$

with $\mathcal{E}_0 = \sum_{\alpha \in E} w_\alpha^0(T^*\mathbb{R}^d)$. Alternatively, if the MTE (3) is written $\partial_t \mathbf{W} = \mathcal{A}\mathbf{W}$ where $\mathcal{A}\mathbf{W} := -\nabla_{\mathbf{z}} \cdot \mathcal{F}(\mathbf{W})$, then its solution in $\mathcal{C}_b^1(T^*\mathbb{R}^d)$ is given by Feynman-Kac's formula :

$$\mathbf{W}(\mathbf{z}, t) = \mathbb{E}_{\mathbf{z}}\{\mathbf{W}^0(\mathbf{Z}_t)\}. \quad (6)$$

This algorithm extends to a bounded domain $\mathcal{D}$ by constructing the trajectories $t \mapsto (\mathbf{z}^j(t), \alpha^j(t))$ from the broken bicharacteristic flow within $T^*\mathcal{D}$, assuming that its intersection with the glancing set remains empty for all times $t \in [0, T]$. This is practically done by reflecting and transmitting the energy rays in $\mathcal{D}$ according to Snell-Descartes laws with all possible mode conversions in E .

3.2 Discontinuous finite element method

3.2.1 Basic principles

Let us now consider the subdivision $\mathcal{T}_h = \cup_{r=1}^{\mathcal{N}} \mathcal{D}_r$ of $\mathcal{D}$ into $\mathcal{N}$ non-overlapping sub-domains, or finite elements $\mathcal{D}_r$. A Galerkin variational formulation of the system (3) in an *ad hoc* functional set $\mathcal{W}_h$ defined on $\mathcal{T}_h$ is: Find $\mathbf{W} \in \mathcal{W}_h$ such that

$$\int_{\mathcal{D}_r \times \mathbb{R}^d} (\partial_t \mathbf{W} \cdot \mathbf{v} - \mathcal{F}(\mathbf{W}) : \nabla_{\mathbf{z}} \mathbf{v}^T) \, d\mathbf{z} + \int_{\partial \mathcal{D}_r \times \mathbb{R}^d} \mathcal{F}(\mathbf{W}) \hat{\mathbf{n}}_r \cdot \mathbf{v} \, d\gamma(\mathbf{z}) = 0, \quad \forall \mathbf{v} \in \mathcal{W}_h, \quad (7)$$

where $\hat{\mathbf{n}}_r$ is outward unit normal to $\mathcal{D}_r$. If the set $\mathcal{W}_h$ is continuous over adjacent subdomains, summing the above expression yields the standard continuous variational formulation:

$$\begin{aligned} \int_{T^*\mathcal{D}} (\partial_t \mathbf{W} \cdot \mathbf{v} - \mathcal{F}(\mathbf{W}) : \nabla_{\mathbf{z}} \mathbf{v}^T) \, d\mathbf{z} + \int_{\Gamma^+(\partial \mathcal{D})} \mathcal{F}(\mathbf{W}) \hat{\mathbf{n}} \cdot \mathbf{v} \, d\gamma(\mathbf{z}) \\ = \int_{\Gamma^-(\partial \mathcal{D})} (\mathbf{f} + \mathcal{R}_{\mathbf{x}}(\mathbf{W})) \cdot \mathbf{v} \, d\gamma(\mathbf{z}), \quad \forall \mathbf{v} \in \mathcal{W}_h. \end{aligned} \quad (8)$$

If however $\mathcal{W}_h$ is discontinuous over two adjacent sub-domains, then the normal flux $\mathcal{F}(\mathbf{W}) \hat{\mathbf{n}}_r$ in Equation (7) has to be replaced by a numerical flux $\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_r$ because the left and right traces of $\mathbf{W}$ on $\partial \mathcal{D}_r$, denoted by $\mathbf{W}^-$ and $\mathbf{W}^+$ respectively, are different *a priori*. The corresponding discontinuous variational formulation is:

$$\int_{\mathcal{D}_r \times \mathbb{R}^d} (\partial_t \mathbf{W} \cdot \mathbf{v} - \mathcal{F}(\mathbf{W}) : \nabla_{\mathbf{z}} \mathbf{v}^T) \, d\mathbf{z} + \int_{\partial \mathcal{D}_r \times \mathbb{R}^d} \mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_r \cdot \mathbf{v} \, d\gamma(\mathbf{z}) = 0, \quad \forall \mathbf{v} \in \mathcal{W}_h, \quad (9)$$

with $\mathbf{W}^\pm = \lim_{h \downarrow 0} \mathbf{W}(\mathbf{x} \pm h \hat{\mathbf{n}}_r, \mathbf{k}, t)$ a.e. $\mathbf{x} \in \partial \mathcal{D}_r$, and the convention:

$$-\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_r = \mathbf{f} + \mathcal{R}_{\mathbf{x}}(\mathbf{W}^-), \quad \mathbf{W}^+ = \mathbf{0}, \quad \mathbf{z} \in \Gamma^-(\partial \mathcal{D}).$$

The numerical flux has also to be consistent, that is $\mathcal{F}_r^*(\mathbf{W}, \mathbf{W}) = \mathcal{F}(\mathbf{W})$, and conservative, that is $\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_r + \mathcal{F}_s^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_s = \mathbf{0}$ if $\partial \mathcal{D}_r \cap \partial \mathcal{D}_s \neq \emptyset$. The

above formulation is the so-called "weak" form of the DG method [15]. The "strong" form corresponding to the original DG method introduced by Lesaint & Raviart [17] is derived by integrating the weak form by parts. Let $\mathcal{L}\mathbf{W} := \partial_t \mathbf{W} + \nabla_{\mathbf{z}} \cdot \mathcal{F}(\mathbf{W})$, then:

$$\int_{\mathcal{D}_r \times \mathbb{R}^d} \mathcal{L}\mathbf{W} \cdot \mathbf{v} \, d\mathbf{z} = \int_{\partial \mathcal{D}_r \times \mathbb{R}^d} (\mathcal{F}(\mathbf{W}^-) - \mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)) \hat{\mathbf{n}}_r \cdot \mathbf{v} \, d\gamma(\mathbf{z}), \quad \forall \mathbf{v} \in \mathcal{W}_h. \quad (10)$$

This last form can also be seen as a penalization method applied to the boundary fluxes of the sub-domains (or elements). An alternative form for linear systems is the ultra-weak variational formulation proposed by Després [10]. It consists in choosing $\mathcal{W}_h = \{\mathbf{v}, \mathcal{L}^* \mathbf{v} = 0\}$, where $\mathcal{L}^* \mathbf{W} \equiv -\partial_t \mathbf{W} - (\mathbf{F}_\alpha \cdot \nabla_{\mathbf{z}} W_\alpha)_{\alpha \in E}$ is the formal adjoint of $\mathcal{L}$. Then (10) is written:

$$\partial_t \left(\int_{\mathcal{D}_r \times \mathbb{R}^d} \mathbf{W} \cdot \mathbf{v} \, d\mathbf{z} \right) + \int_{\partial \mathcal{D}_r \times \mathbb{R}^d} \mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_r \cdot \mathbf{v} \, d\gamma(\mathbf{z}) = 0, \quad \forall \mathbf{v} \in \mathcal{W}_h. \quad (11)$$

The test functions $\mathbf{v}$ have then to be taken as solutions of an adjoint problem, typically plane waves in piecewise homogeneous media [12]: $\mathbf{v}(\mathbf{z}, t) = \mathbf{V} \phi(\mathbf{k} \cdot \mathbf{x} - \omega t)$ such that $\Gamma(\mathbf{k})\mathbf{V} = \omega \mathbf{V}$, meaning that ω and $\mathbf{V}$ are the eigenvalues and eigenvectors of the acoustic tensor Γ which is simply $\Gamma(\mathbf{k}) = \text{diag}(\lambda_\alpha(\mathbf{x}, \mathbf{k}))_{\alpha \in E}$ here.

3.2.2 Numerical implementation

The numerical implementation of the DG method expounded above requires three main ingredients, once the subdivision $\mathcal{T}_h$ has been performed: (i) the definition of numerical fluxes $\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+)$ on all elements, (ii) the choice of the approximation set $\mathcal{W}_h$, and (iii) the choice of a time-integration scheme. The computational domain in phase-space is reduced to the unit cosphere bundle $S^* \mathcal{D} = \mathcal{D} \times \mathbb{S}^{d-1}$ by considering only normalized wave vectors $\hat{\mathbf{k}}$ defined on the unit sphere $\mathbb{S}^{d-1}$ of $\mathbb{R}^d$. For applications to assemblies of slender structures, $d = 1$ (beams) and $\mathbb{S}^0 = \{-1, +1\}$, or $d = 2$ (shells) and the unit circle $\mathbb{S}^1$ is parametrized by $\theta \in [0, 2\pi[$ such that $\hat{\mathbf{k}} = (\cos \theta, \sin \theta)^T$. In both cases the meshing of $S^* \mathcal{D}$ writes $\mathcal{T}_h = \cup_{r=1}^N X_r = \cup_{r=1}^N \mathcal{D}_r \times \mathbb{S}^{d-1}$, denoting by h the largest diameter of the elements $\mathcal{D}_r$.

Regarding (i), the numerical fluxes ensuring the stability and consistency of the DG method for transport equations have the form [5]:

$$\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) = \frac{1}{2} (\mathcal{F}(\mathbf{W}^-) + \mathcal{F}(\mathbf{W}^+)) + \mathbf{C}_r (\mathbf{W}^- - \mathbf{W}^+) \hat{\mathbf{n}}_r$$

where $\mathbf{C}_r$ is an $M \times M$ non-negative definite matrix. For example, the (upwind) Roe flux corresponds to $\mathbf{C}_r = \frac{1}{2} \text{diag}(c_\alpha |\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_r|)_{\alpha \in E}$ and the Lax-Friedrichs flux corresponds to $\mathbf{C}_r = \frac{1}{2} \text{diag}(c_\alpha)_{\alpha \in E}$. This expression can be generalized as follows for an element boundary $\partial \mathcal{D}_r$ belonging to an interface (a junction) γ_D where the group

velocities c_α are discontinuous. Let $\gamma_{Dr} = \partial\mathcal{D}_r \cap \gamma_D \neq \emptyset$, then having in mind Equation (5), the numerical flux is written on $\Gamma^+(\gamma_{Dr})$:

$$\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_r = \left(\frac{1}{2} - C_r\right) \mathcal{F}(\mathbf{W}^-) \hat{\mathbf{n}}_r + \left(\frac{1}{2} + C_r\right) (\mathcal{R}_D^+(\mathbf{W}^+) + \mathcal{T}_D^-(\mathbf{W}^-)) \quad (12)$$

and on $\Gamma^-(\gamma_{Dr})$:

$$\mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) \hat{\mathbf{n}}_r = \left(\frac{1}{2} - C_r\right) \mathcal{F}(\mathbf{W}^+) \hat{\mathbf{n}}_r - \left(\frac{1}{2} + C_r\right) (\mathcal{R}_D^-(\mathbf{W}^-) + \mathcal{T}_D^+(\mathbf{W}^+)) \quad (13)$$

with $0 \leq |C_r| \leq \frac{1}{2}$. This definition extends in fact to the case $\partial\mathcal{D}_r \cap \gamma_D = \emptyset$ for which $\mathcal{R}_D^\pm(\mathbf{W}^\pm) = 0$ and $\mathcal{T}_D^\pm(\mathbf{W}^\pm) = \mp \mathcal{F}(\mathbf{W}^\mp) \hat{\mathbf{n}}_D$ on $\Gamma^\mp(\partial\mathcal{D}_r)$:

$$\begin{aligned} \mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) &= \left(\frac{1}{2} - C_r\right) \mathcal{F}(\mathbf{W}^-) + \left(\frac{1}{2} + C_r\right) \mathcal{F}(\mathbf{W}^+), \quad \mathbf{z} \in \Gamma^+(\partial\mathcal{D}_r), \\ \mathcal{F}_r^*(\mathbf{W}^-, \mathbf{W}^+) &= \left(\frac{1}{2} - C_r\right) \mathcal{F}(\mathbf{W}^+) + \left(\frac{1}{2} + C_r\right) \mathcal{F}(\mathbf{W}^-), \quad \mathbf{z} \in \Gamma^-(\partial\mathcal{D}_r). \end{aligned}$$

Regarding (ii), the approximation set $\mathcal{W}_h \subset L^2(\mathcal{S}^*\mathcal{D})$ is constructed by tensoring the physical space and phase variables $\mathcal{W}_h = \mathcal{D}_h \otimes \mathcal{S}_h$, with :

$$\mathcal{D}_h = \{\mathbf{v} \in (L^2(\mathcal{D}))^d; v_\alpha|_{\mathcal{D}_r} \in \mathbb{P}_p(\mathcal{D}_r), 1 \leq r \leq \mathcal{N}, \alpha \in E\}$$

where $\mathbb{P}_p(\mathcal{D}_r) \subset H^1(\mathcal{D}_r)$ is typically a polynomial set, and $\mathcal{S}_h$ is a finite dimensional set of real-valued functions defined on $\mathbb{S}^{d-1}$. Practically in dimension 1, $\mathbb{P}_p(\mathcal{D}_r)$ is spanned by either a set of Jacobi orthogonal polynomials up to the order p (the modal, or spectral approach), or a set of Lagrange interpolation polynomials associated to $p + 1$ interpolation points (the nodal approach). The degrees of freedom in physical space are the projection coefficients on the Jacobi polynomials of the approximate solution in $\mathcal{W}_h$ in the modal approach, and its values at the interpolation points in the nodal approach. It should be noted that if the latter are chosen as Gauss-Jacobi quadrature points on $[-1, 1]$, the corresponding Lagrange polynomials are computed explicitly as functions of Jacobi polynomials, which yields a unified framework for the modal and nodal approaches. The numerical and dispersion errors introduced by this spatial discretization are $O(K^{2p+3})$ and $O(K^{2p+2})$, respectively, for decentered numerical fluxes ($C_r \neq 0$), or $O(K^{4[\frac{p}{2}] + 3})$ and 0 for centered fluxes ($C_r = 0$) [1], where $K = h|\mathbf{k}|$ and $[\cdot]$ stands for the integer part. These so-called "superconvergence" properties of DG methods are particularly attractive for long-times numerical simulations. The spatial discretization in dimension 2 is achieved by tensoring both dimensions on structured meshes, as proposed in [7] for example. The construction of $\mathcal{S}_h$ is however more involved. The modal approach corresponds to a truncated Fourier expansion in θ , but this scheme does not ensure the positivity of the approximate solution because of the slowness of the convergence of Fourier series. The nodal

approach by Lagrange interpolation yields the well-know ray effect which also arises in the discrete ordinate methods widely used in neutronic or thermal transfer computations [11, chap. 8]. Therefore further researches are needed on this issue; it is the subject of ongoing investigations.

At last, regarding (iii), strong stability-preserving (SSP), high-order Runge-Kutta schemes have been implemented [14]. The CFL condition of these schemes, coupled to the above mentioned phase-space discretization methods, can be computed numerically for various orders p .

4 Numerical examples

In the examples below the transport equations (1) with specular-like boundary and interface (at the junctions) conditions are solved numerically by the schemes expounded in the foregoing section. In Sect. 4.1 the example of [26] of scalar transport in coupled Timoshenko beams is resumed and extended, whereas Sect. 4.2 deals with two coupled thick cylindrical shells.

4.1 Coupled beams

We first consider an assembly of two Timoshenko beams $] - L, 0[\cup] 0, L[, L > 0$, with a junction at $x = 0$ such that beam #2 forms an angle ϕ with beam #1. Three types of energy rays propagate in such a system: a pure shear transverse mode at the velocities c_{Tj} ($j = 1$ or 2 for either beam #1 or beam #2) denoted by subscript T, a pure bending mode at the velocities c_{Pj} denoted by subscript Pb, and a longitudinal mode at the velocities c_{Pj} denoted by subscript Pn; see [24, 26] for some details on the derivation. Therefore the energy density evolution in this system is depicted by three coupled, one-dimensional transport equations ($M = 3$ and $d = 1$ such that $\hat{k} \equiv \text{sign}(k)$) for $\alpha = \text{T, Pb or Pn}$ written as:

$$\partial_t w_\alpha + c_\alpha \hat{k} \partial_x w_\alpha + \eta_\alpha(k) w_\alpha = 0, \quad (14)$$

where the coupling is ensured by the left ($-$) and right ($+$) power reflection and transmission operators $\mathcal{R}_D^\pm$, $\mathcal{T}_D^\pm$ of the junction. The expressions of the kernels of $\mathcal{R}_D^\pm$ and $\mathcal{T}_D^\pm$ as functions of ϕ and beams mechanical and geometrical characteristics are derived in [26], using a high-frequency model compatible with the propagation characteristics obtained by the transport theory. This study has also shown that the power reflection coefficients at a beam end for either Dirichlet (clamped) or Neumann (free) boundary condition were all equal to 1, without mode conversion of any type; therefore specular reflections are considered at the boundaries $x = \pm L$. All computations are performed for the initial conditions:

$$w_T^0(x, k) = \varphi \left(\frac{x - x_0}{r_0} \right) \otimes \delta(\hat{k} - \hat{k}_0), \quad w_{\text{Pb}}^0 = w_{\text{Pn}}^0 \equiv 0,$$

with $\varphi(x) = \max(0, 1 - 2|x|)$ ("hat" source) or $\varphi(x) = \mathbf{1}_{\{|x| \leq \frac{1}{2}\}}(x)$ ("square" source). The parameters are chosen as $x_0 = -0.1 \times L$, $r_0 = 0.12 \times L$. Beam velocities are such that $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, their Poisson's ratio being $\nu_1 = \nu_2 = 0.3$, and the junction angle is $\phi = \pi/3$.

The coupled transport equations (14) are solved numerically by the Runge-Kutta DG finite element method expounded in Sect. 3.2 above. The assembly is partitioned into 40 uniform spatial elements, and Eqs. (14) are solved in time by a eight-order SSP Runge-Kutta scheme. The Courant number for the simulations is fixed at $c \equiv c_{T2} \frac{\Delta t}{\Delta x} = 2.5 \cdot 10^{-2}$. Figure 1 and Figure 2 display our results for either a "hat" source with $k_0 = +1$ and 10-th order Legendre polynomial bases, or a "square" source with $k_0 = -1$ and 10-th order Lagrange polynomial bases, using Gauss-Lobatto interpolation points. In this latter simulation, the reconstructed numerical solution has been filtered in order to tame the Gibbs phenomenon [29]. The time scale is $T = L/c_{T2}$, the time needed by the wave front to reach the extremity $x = L$ of the homogeneous assembly starting from the junction. The thin vertical line indicates the location of the junction. The wavefronts of shear ($\alpha = T$) and bending/axial ($\alpha = P$) energy densities propagating at the speeds c_{Tj} and c_{Pj} , respectively, are marked by small capital letters on Figure 1. A small amount of damping $\eta_\alpha = 0.002 \text{ s}^{-1}$ has been added such that the total energy within the system at a fixed wavenumber k evolves as (see Equation (2)):

$$\mathcal{E}(t) = \sum_{\alpha=T,Pb,Pn} e^{-\eta_\alpha t} \mathcal{E}_\alpha^0, \quad \mathcal{E}_\alpha^0 = \int_{-L}^L w_\alpha^0(x, \pm|k|) dx. \quad (15)$$

This exact result is recovered numerically by our scheme, as shown by Figure 3 and Figure 4 for both types of sources and discretizations. They display the overall mechanical energies $\mathcal{E}_j^h(t)$ computed within each beam by $\mathcal{E}_1^h(t) = \int_{-L}^0 \mathcal{E}^h(x, t) dx$, $\mathcal{E}_2^h(t) = \int_0^L \mathcal{E}^h(x, t) dx$, and their sum, normalized by the total energy introduced into the system $\mathcal{E}(0) = \sum_\alpha \mathcal{E}_\alpha^0 = \mathcal{E}_T^0$. The computed and exact total energies perfectly match on these plots: one concludes that the proposed scheme in dimension one is numerically conservative.

These results exhibit the following features: one observes a beating phenomenon of the total energy within each subsystem, by which it is transferred back and forth from the first beam to the second one and the other way round. At late time an equipartition regime holds between both beams. It is described by diffusion equations [8, 28]. The onset of diffusion is prompted by the mixing of wave fronts and modes at the junction, and to a lesser extent at the boundaries. This mixing process is accelerated by the presence of random heterogeneities, as outlined in [27]. Finally, the long-time limit of the ratio $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Tj}^h(t)$ of the bending/longitudinal energy over the shear energy within each beam is expected to be the ratio c_{Tj}/c_{Pj} to the power of the physical dimension ($d = 1$ in the present case) independently of the source type, in accordance with the results of [20, 25] for example. Here this universal equilibration rule is recovered by the DG scheme, as exemplified by Figure 5 and Figure 6 for both sources.

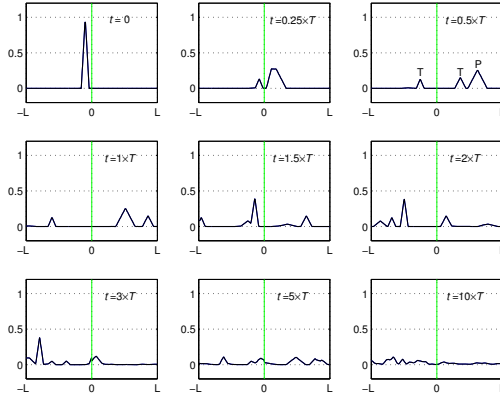


Figure 1: Evolution of the energy density in an assembly of two Timoshenko beams for $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.002 \text{ s}^{-1}$, $T = L/c_{T2}$, and a "hat" source with $\hat{k}_0 = +1$. 8-th order SSP Runge-Kutta DG method with 10-th order Legendre modal expansion and $\mathcal{N} = 40$ elements.

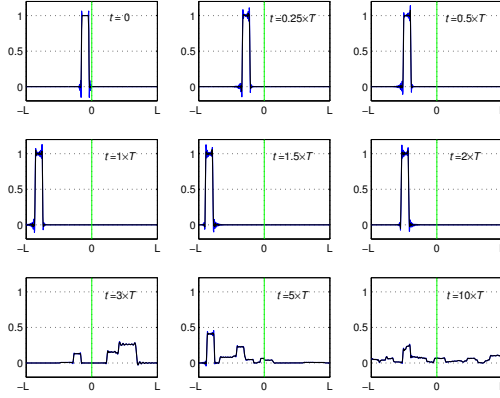


Figure 2: Evolution of the energy density in an assembly of two Timoshenko beams for $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.002 \text{ s}^{-1}$, $T = L/c_{T2}$, and a "square" source with $\hat{k}_0 = -1$. 8-th order SSP Runge-Kutta DG method with 10-th order Gauss-Lobatto nodal expansion and $\mathcal{N} = 40$ elements, — unfiltered solution, — exponential filter.

4.2 Coupled shells

Next we consider an assembly of two thick cylindrical shells, the first one with a constant radius (shell #1) and the second one with a variable radius (shell #2). Both cylinders have the same length L which is also twice the constant radius of the first shell. The angle between the shells is again denoted by ϕ , and their mean surfaces are denoted by $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively. Five types of energy rays propagate in such a system: a pure shear transverse mode at the velocities c_{Tj} ($j = 1$ or 2 for either shell #1 or shell #2) denoted by subscript T, and coupled longitudinal and transverse modes P and S each of multiplicity 2 corresponding to either in-plane (subscript n) or bending energies (subscript b), at the velocities c_{Pj} and c_{Sj} respectively; see [23] for some de-

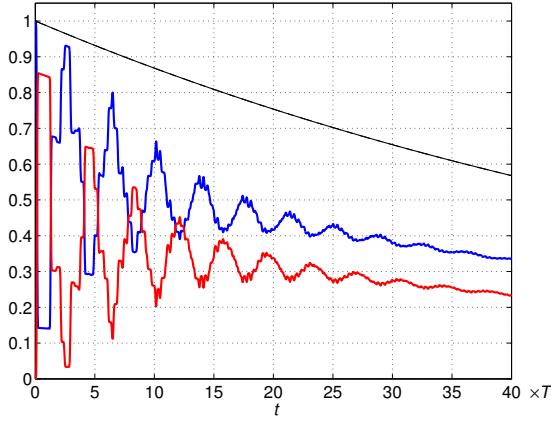


Figure 3: Evolution of the total vibrational energies in an assembly of two Timoshenko beams for $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.002 \text{ s}^{-1}$, $T = L/c_{T2}$, and a "hat" source with $\hat{k}_0 = +1$. 8-th order SSP Runge-Kutta DG method with 10-th order Legendre modal expansion and $\mathcal{N} = 40$ elements, — beam #1, — beam #2, — computed and exact total energy.

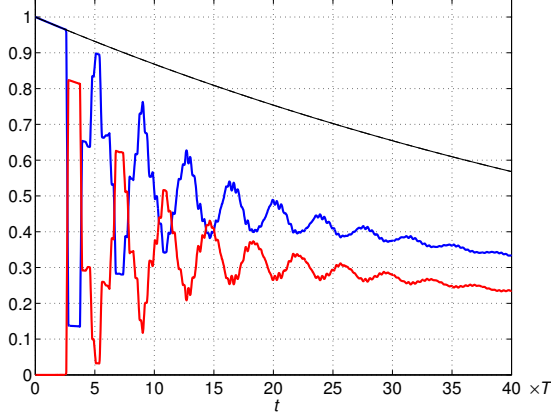


Figure 4: Evolution of the total vibrational energies in an assembly of two Timoshenko beams for $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.002 \text{ s}^{-1}$, $T = L/c_{T2}$, and a "square" source with $\hat{k}_0 = -1$. 8-th order SSP Runge-Kutta DG method with 10-th order Gauss-Lobatto nodal expansion and $\mathcal{N} = 40$ elements, — beam #1, — beam #2, — computed and exact total energy.

tails on the derivation. Then the energy density evolution in this system is depicted by five coupled transport equations (1) with $M = 5$ —since $\alpha = \text{T, Pb, Pn, Sb or Sn}$ —and $d = 2$, where the coupling is ensured by the left (−) and right (+) power reflection and transmission operators $\mathcal{R}_D^\pm$, $\mathcal{T}_D^\pm$ of the shell junction. Their expressions as functions of ϕ , the incident tangent wave vector $\mathbf{k}_t = (\mathbf{I}_2 - \hat{\mathbf{n}}(\mathbf{x}_t) \otimes \hat{\mathbf{n}}(\mathbf{x}_t))\mathbf{k}$ on the junction line $\gamma_D \ni \mathbf{x}_t$, and shells mechanical and geometrical characteristics are derived in [26], using a high-frequency model compatible with the propagation characteristics obtained by the transport theory within plates. As the curvature tensor has no influence on the

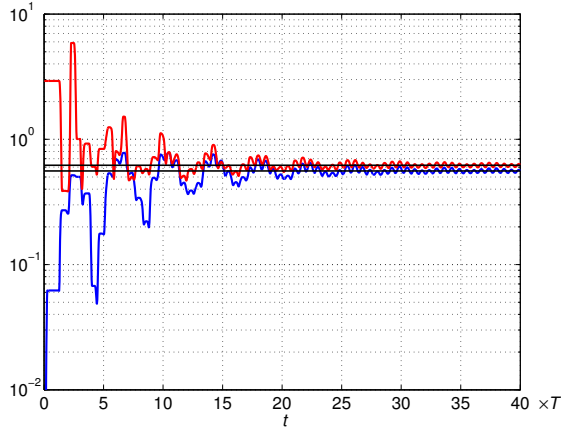


Figure 5: Evolution of the ratio $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Tj}^h(t)$ in an assembly of two Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.002 \text{ s}^{-1}$, $T = L/c_{T2}$, and a "hat" source with $\hat{k}_0 = +1$. 4-th order SSP Runge-Kutta DG method with 5-th order Legendre modal expansion and $\mathcal{N} = 40$ elements, — beam #1 ($j = 1$), — beam #2 ($j = 2$), — c_{Tj}/c_{Pj} .

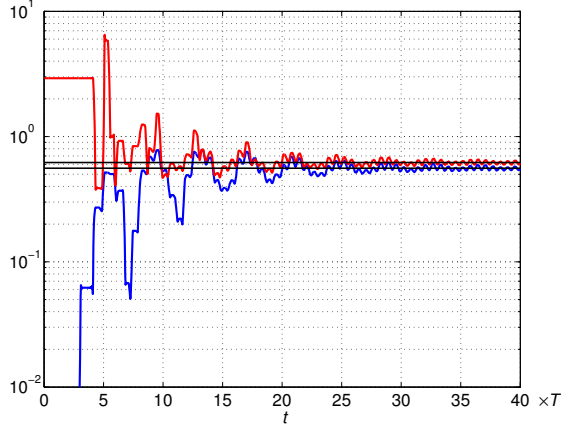


Figure 6: Evolution of the ratio $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Tj}^h(t)$ in an assembly of two Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $\nu_1 = \nu_2 = 0.3$, $\eta = 0.002 \text{ s}^{-1}$, $T = L/c_{T2}$, and a "square" source with $\hat{k}_0 = -1$. 4-th order SSP Runge-Kutta DG method with 5-th order Gauss-Lobatto nodal expansion and $\mathcal{N} = 40$ elements, — beam #1 ($j = 1$), — beam #2 ($j = 2$), — c_{Tj}/c_{Pj} .

high-frequency regime in this model, the same reflection/transmission kernels as for flat plates are used in this example. Neumann boundary conditions are considered at the extremities of the shells, and the corresponding power reflection coefficients have been derived in [26] as well. Here it has been shown that the shear mode is uncoupled to the in-plane and bending modes by boundary reflections, however the longitudinal P and transverse S modes are coupled. Otherwise all these modes are fully coupled by the junction. The initial condition is an isotropic shear impact $w_T^0(\mathbf{x}, \mathbf{k}) = \delta(\mathbf{x} - \mathbf{x}_0)$,

independently of $\mathbf{k}$, all other modes being initially unloaded. Here $\mathbf{x}_0 = (-L/2, 0)$ and the junction line is the circle $\gamma_D = \{\mathbf{x} = (0, y), 0 \leq y \leq \pi L\}$ in curvilinear coordinates. The Young's moduli, densities, and thicknesses of both shells are equal, their Poisson's ratios are $\nu_1 = 0.3$ and $\nu_2 = 0.2$, and the junction angle is $\phi = \pi/12$.

The coupled transport equations (1) are solved in this case by a direct Monte-Carlo method [16]. The computer code developed for this application has been fully vectorized, in such a way that numerical simulations of $O(10^6)$ ray paths with $O(10^2)$ time steps can be performed in less than one hour of CPU time on a personal computer. Figure 7 through Figure 9 below display our results with the time scale $T = L/c_{T2}$. A small amount of damping $\eta = 0.002 \text{ s}^{-1}$ has been added as for the beam assembly of the previous example. Figure 7 displays the evolution of the energy density within the shell assembly, while Figure 8 displays the evolution of the overall mechanical energies within each shell and their sum; it shows that the Monte-Carlo scheme is numerically conservative. In addition Figure 9 displays the ratios $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Sj}^h(t)$ of the total bending/longitudinal and transverse energies within each shell, respectively, where:

$$\mathcal{E}_{Pj}^h(t) = \sum_{\alpha=\text{Pn,Pb}} \int_{\mathcal{D}_j \times S^1} w_{\alpha}^h(\mathbf{x}, \mathbf{k}, t) d\mathbf{x} d\mu(\hat{\mathbf{k}}), \quad j = 1, 2,$$

and a similar expression for $\mathcal{E}_{Sj}^h(t)$. All computations exhibit the same features as for the assembly of beams, calling for similar qualitative comments. The diffusive limit at large times of $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Sj}^h(t)$ is expected to be the ratio c_{Sj}/c_{Pj} to the power of the physical dimension $d = 2$ independently of the source type [25]. This universal equilibration rule is recovered by the Monte-Carlo scheme as well, see Figure 9. It should be reminded, however, that these results ignore the contribution of the glancing rays possibly trapped by the boundaries and the junction line, because a theoretical model lacks at present to describe them. Indeed it is known from experiments that the vibrational energy is very much likely to be guided by the interfaces, stiffeners or junctions in a real-life complex structure, so these effects should not be disregarded. Further developments are needed on this issue.

5 Conclusions

A transport model for the vibrational energy evolution in heterogeneous slender structures has been proposed. The transport equations have been solved numerically by a discontinuous finite element method and a Monte-Carlo method, among other possible schemes. Both methods have been implemented and tested on examples of assemblies of thick beams or cylindrical shells. Direct numerical simulations illustrate the transport regime for these systems, and the onset of a diffusive regime at late times. The latter is characterized by energy equilibration rules very much comparable to the assumption of modal equipartition invoked in the statistical energy analysis of structural-acoustics systems [18]. However a rigorous mathematical model of the diffusive regime for bounded elastic media lacks at present. It should be able to predict

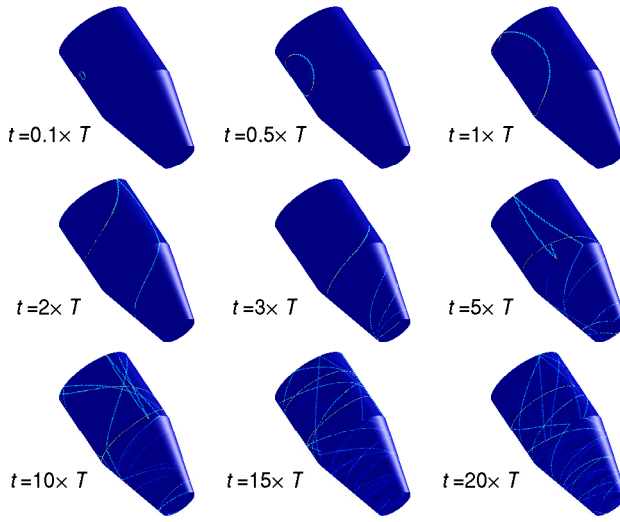


Figure 7: Evolution of the energy density in an assembly of two cylindrical shells for $\phi = \frac{\pi}{12}$, $\nu_1 = 0.3$, $\nu_2 = 0.2$, $\eta = 0.002 \text{ s}^{-1}$, and $T = L/c_{T2}$. Monte-Carlo method with $N = 10^6$ path samples.

the equilibration ratios as function of the sub-systems mechanical parameters. Another open issue is the consideration of the glancing set of energy rays—the diffractive and gliding ones—for the part of the power flows possibly trapped by the interfaces and junctions. Both aspects are the subject of ongoing research.

Acknowledgment

The author acknowledges the support of the French National Research Agency (contract n. ANR-06-CIS6-003) for some parts of this work.

References

- [1] M. Ainsworth, "Dispersive and dissipative behaviour of high order discontinuous Galerkin finite element methods", *Journal of Computational Physics*, 198(1), 106–130, 2004.
- [2] J.-L. Akian, "Semiclassical measures for tridimensional elastodynamics: boundary conditions for the hyperbolic set", *Asymptotic Analysis*, submitted, 2010.
- [3] G. Bal, "Kinetics of scalar wave fields in random media", *Wave Motion*, 43(2), 132–157, 2005.
- [4] A. Bensoussan, J.-L. Lions, and G. C. Papanicolaou, "Boundary layers and homogenization of transport processes", *Publications of the Research Institute for Mathematical Sciences, Kyoto University*, 15, 53–157, 1979.

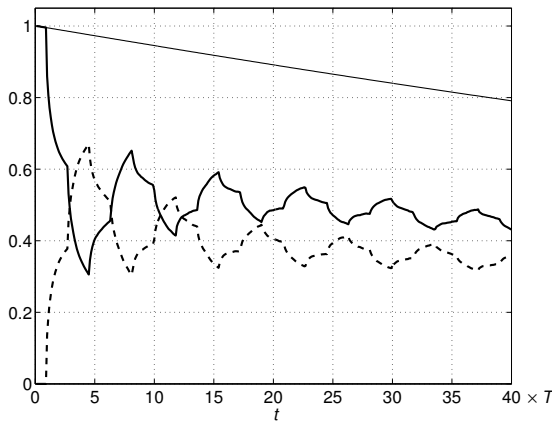


Figure 8: Evolution of the total vibrational energies in an assembly of two cylindrical shells for $\phi = \frac{\pi}{12}$, $\nu_1 = 0.3$, $\nu_2 = 0.3$, $\eta = 0.002 \text{ s}^{-1}$, and $T = L/c_{T2}$. Monte-Carlo method with $N = 10^6$ path samples.

- [5] F. Brezzi, L. D. Marini, and E. Süli, "Discontinuous Galerkin methods for first-order hyperbolic problems", *Mathematical Models and Methods in Applied Sciences*, 14(12), 1893–1903, 2004.
- [6] B. Cockburn, G. E. Karniadakis, and C.-W. Shu, editors, "Discontinuous Galerkin Methods. Theory, Computation and Applications", Springer-Verlag, Berlin, 2000.
- [7] G. Cohen, X. Ferrières, and S. Pernet, "A spatial high-order hexahedral discontinuous Galerkin method to solve Maxwell's equations in time domain", *Journal of Computational Physics*, 217(2), 340–363, 2006.
- [8] C. Costantini, T. G. Kurtz, "Diffusion approximation for transport processes with general reflection boundary conditions", *Mathematical Models and Methods in Applied Sciences*, 16(5), 717–762, 2006.
- [9] R. Dautray and J.-L. Lions, "Mathematical Analysis and Numerical Methods for Science and Technology", volume 6, Springer, Berlin, 1993.
- [10] B. Després, "Sur une formulation variationnelle de type ultra-faible", *Comptes Rendus de l'Académie des Sciences Série I*, 318, 939–944, 1994.
- [11] J. J. Duderstadt and W. R. Martin, "Transport Theory", John Wiley & Sons, New York, 1979.
- [12] G. Gabard, "Discontinuous Galerkin methods with plane waves for time-harmonic problems", *Journal of Computational Physics*, 225(2), 1961–1984, 2007.
- [13] P. Gérard, P. A. Markowich, N. J. Mauser, and F. Poupaud, "Homogenization limits and Wigner transforms", *Communications on Pure and Applied Mathematics*, L(4), 323–379, 1997.
- [14] S. Gottlieb, C.-W. Shu, and E. Tadmor, "Strong stability-preserving high-order time discretization methods", *SIAM Review*, 43(1), 89–112, 2001.
- [15] J. S. Hesthaven and T. Warburton, "Nodal Discontinuous Galerkin Methods", Springer, New York, 2008.

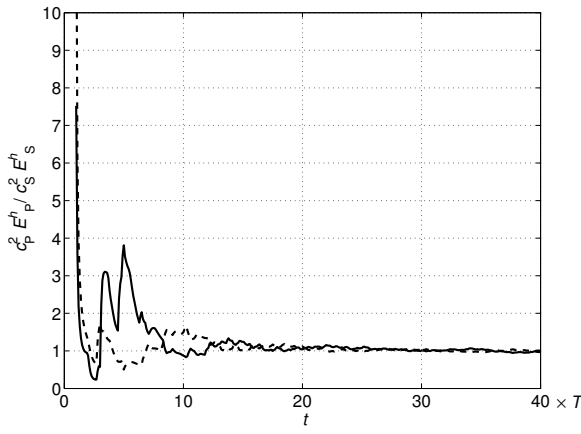


Figure 9: Evolution of the ratio $c_{Pj}^2 \mathcal{E}_{Pj}^h(t) / c_{Sj}^2 \mathcal{E}_{Sj}^h(t)$ of the total bending/longitudinal and transverse energies in an assembly of two cylindrical shells for $\phi = \frac{\pi}{12}$, $\nu_1 = 0.3$, $\nu_2 = 0.2$, $\eta = 0.002 \text{ s}^{-1}$, and $T = L/c_{T2}$. Solid line: shell #1 ($j = 1$), dashed line: shell #2 ($j = 2$). Monte-Carlo method with $N = 10^6$ path samples.

- [16] B. Lapeyre, É. Pardoux, and R. Sentis, "Introduction to Monte-Carlo Methods for Transport and Diffusion Equations", Oxford University Press, Oxford, 2003.
- [17] P. Lesaint and P.-A. Raviart, "On a finite element method for solving the neutron transport equation", In C. De Boor, editor, *Mathematical Aspects of Finite Elements in Partial Differential Equations*, pages 89–123. Mathematics Research Center, the University of Wisconsin, Madison WI, Academic Press, New York, 1974.
- [18] R. H. Lyon and R. G. DeJong, "Theory and Application of Statistical Energy Analysis", Butterworth-Heinemann, Boston, 2nd edition, 1995.
- [19] D. J. Nefske and S. H. Sung, "Power flow finite element analysis of dynamic systems: basic theory and application to beams", *ASME Journal of Vibration, Acoustics, Stress and Reliability in Design*, 111(1), 94–100, 1989.
- [20] G. C. Papanicolaou and L. V. Ryzhik, "Waves and transport", In L. Caffarelli and W. E, editors, *Hyperbolic Equations and Frequency Interactions*, volume 5 of *IAS/Park City Mathematics Series*, pages 305–382. American Mathematical Society, Providence, 1999.
- [21] W. H. Reed and T. R. Hill, "Triangular mesh methods for the neutron transport equation", Technical Report LA-UR-73-479, Los Alamos Scientific Laboratory, Los Alamos, 1973.
- [22] É. Savin, "Midfrequency vibrations of a complex structure: Experiments and comparison with numerical simulations", *AIAA Journal*, 40(9), 1876–1884, 2002.
- [23] É. Savin, "Transient transport equations for high-frequency power flow in heterogeneous cylindrical shells", *Waves in Random Media*, 14(3), 303–325, 2004.
- [24] É. Savin, "High-frequency vibrational power flows in randomly heterogeneous structures", In G. Augusti, G. I. Schuëller, and M. Ciampoli, editors, *Proceedings of the 9th International Conference on Structural Safety and Reliability ICOS-*

- SAR 2005, Rome, 19-23 june 2005, pages 2467–2474. Millpress Science Publishers, Rotterdam, 2005.
- [25] É. Savin, "Derivation of diffusion equations for high-frequency vibrations of randomly heterogeneous structures", In B. H. V. Topping, G. Montero, and R. Montenegro, editors, *Proceedings of the Eighth International Conference on Computational Structures Technology*, Las Palmas de Gran Canaria, 12-15 September 2006, paper #238. Civil-Comp Press, Stirlingshire, 2006. doi:10.4203/ccp.83.238
- [26] É. Savin, "A transport model for high-frequency vibrational power flows in coupled heterogeneous structures", *Interaction and Multiscale Mechanics*, 1(1), 53–81, 2007.
- [27] É. Savin, "High-frequency vibrational power flows in randomly heterogeneous structures", In A. K. Belyaev and R. S. Langley, editors, *Vibration Analysis of Structures with Uncertainties*, IUTAM Bookseries, in press. Springer, Berlin, 2010.
- [28] R. Sentis, "Équations de transport avec conditions aux limites de type réflexion", Research Report 162, INRIA, Rocquencourt, september 1982.
- [29] H. Vandeven, "Family of spectral filters for discontinuous problems", *Journal of Scientific Computing*, 6(2), 159–192, 1991.